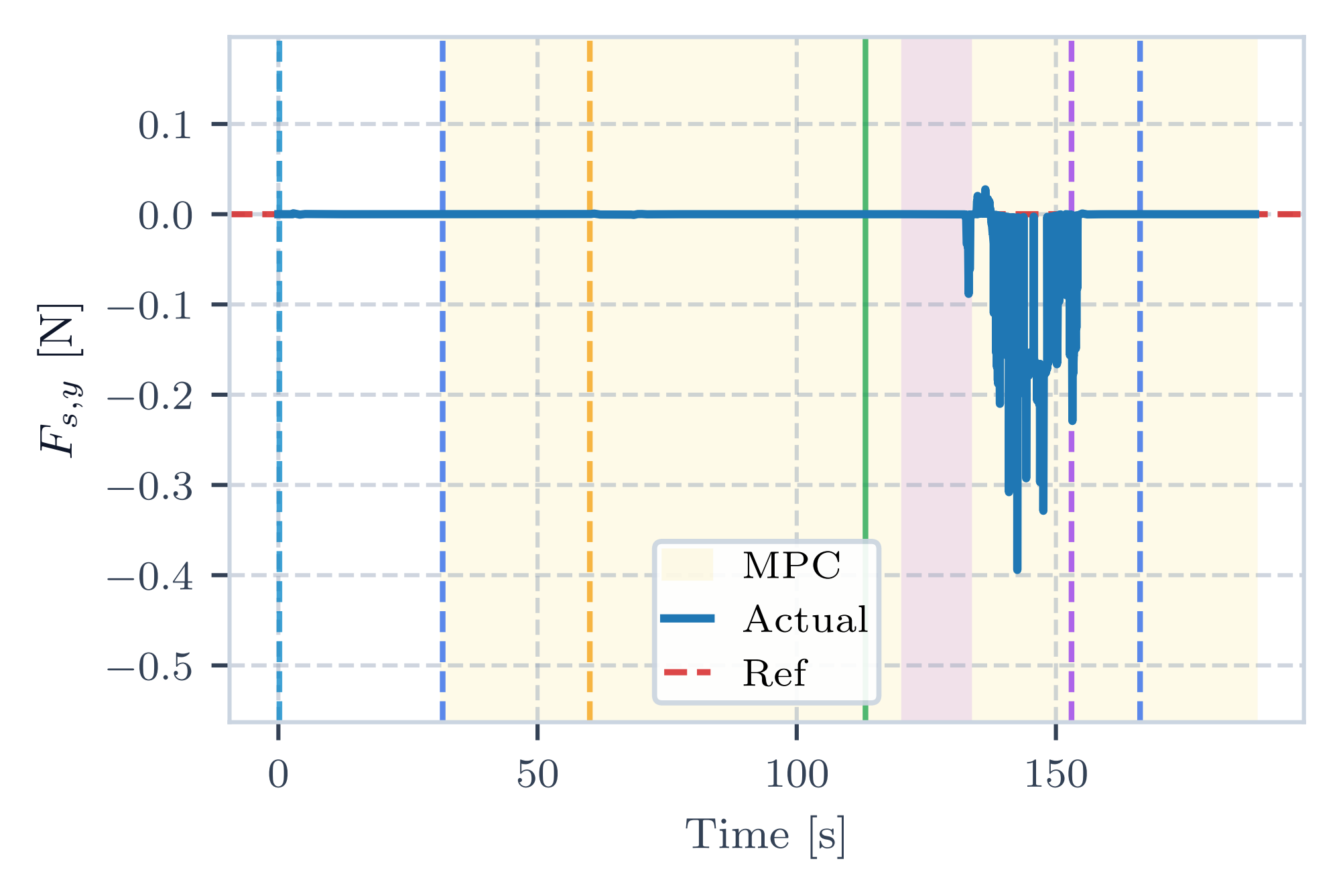

0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
$F_{s,y}$ [N]
0
50
100
150
Time [s]
MPC
Actual
Ref